MB cache, which was bettered only by the massive 300 GB Maxtor One Touch II, which had 16 MB of cache.

The drive that came second here, and second by a long way from the third-placed Freecom drive, was the Western Digital 40 GB Passport. The drive had fabulous 31 MBps sequential read and 21 MBps random read speeds.

The performance of the other drives was strictly ordinary, though special mention must be made of the 2.5 and 5 GB Seagate pocket drives, for the simple reason that they're so small. They utilised 1-inch drives as opposed to 1.8 inches for the Transcend drives and 2.5 inches for the Freecom, WD and Seagate 100 GB.

Sadly though, these drives were our worst performers, with speeds of 6 and 8 MBps respectively for the sequential read and sequential write tests. They have a spindle speed of 3,600 rpm, whereas all the other drives were at least 4,200 rpm. In fact, the Seagate 100 GB and WD Passport were 5,400 rpm drives. Remember, the faster your drive spins, the better the transfer rate and the lower the seek times.

Seagate 2.5 GB Pocket Hard Drive

CPU Utilisation: There is nothing much to differentiate these drives in terms of CPU utilisation—they just varied between 3.4 per cent for the Seagate 2.5 GB to 3.75 for the Seagate 100 GB. This was true even for our tests in the desktop storage/backup category, so there really isn't too much point dwelling on this.

So why do we need to talk about CPU utilisation at all? There are other drives out there in the market that hog much more than 3.75 per cent, but in the batch of drives we received for this shootout, it just so happened that no drive—in either the portable or the desktop category—was worse than roughly 4 per cent.

Real-World Tests: We now come to our real-world test results. For these tests, we used a single 1 GB file for our sequential read/write tests, and an assorted set of MP3 files totalling 1 GB. The files were copied to and from the hard disk, and we recorded time using a stopwatch.

Here, 'write' means copying files to the hard drive being tested, and 'read' means copying files from the disk.

Transcend Storejet 40GB

The results we saw were mixed. There was no drive that beat all the others in every test. But the drive that did beat the others in most of the tests was the Seagate 100 GB. It was beaten fair and square in two tests (and that's half of the real-world tests!)—the assorted read test—where it was beaten by the WD Passport, which clocked a time of 39.95 as compared to 40.26 seconds for the Seagate—and, rather surprisingly, the sequential write test. Here it was beaten by the Freecom FHD 2 Pro 60 GB, which clocked 42.89 seconds as compared to 47.02 for the Seagate.

The worst performers were again the Seagate 2.5 and 5 GB disks, with times that were as much as 5 times slower than the frontrunners.

The drive that really impressed, apart from the lightning-fast Seagate 100 GB, was the WD Passport. It almost matched the Seagate in every test and beat it in one, and was light-years ahead of the rest of the pack.

The Transcend drives were slow considering they have the same spindle speed and cache as the Freecom drive. This shows that the Freecom disk's cache

How We Tested

Our test bed comprised an AMD Athlon 64 FX 53 Processor, with an MSI K8T Neo 2 motherboard, 512 MB of dual-channel DDR RAM, along with a Seagate 120 GB SATA HDD. The graphics card we used was a GeForce 5700 LE (128 MB).

The hard disk was formatted as FAT 32 and loaded with Windows XP Professional with Service Pack 1.

The drives were benchmarked on the basis of Features, Performance and Price.

We ran several benchmarks on the drives to find out which was the best performing, and then weighed that against the price. We were looking for the drive with the best all-round performance, features, as well as price.

Features

We looked at the various inbuilt characteristics and qualities of each of the drives. For the products being marketed as portable devices, we laid particular emphasis on the drive's weight and dimensions for 1.8 and 2.5-inch drives only. We looked less at the weight and dimensions of the 3.5-inch drives, because they will be primarily used as a stationary backup device.

We also observed the build quality, with more importance being given to the sturdiness of the product. Extra points were awarded for features such as extra USB/FireWire ports, configurable buttons, and so on.

Performance

To gauge performance, we conducted two types of tests-synthetic and real-world.

For the synthetic tests, we used the SiSoft Sandra Professional 2005 benchmarking suite to assess theoretical read and write speeds, as well as average access times. We also tested with Winbench 99 2.0 to measure the CPU utilisation of each drive.

In the real-world tests, we transferred a set of assorted files of varying sizes totalling 1 GB, back and forth from the system hard disk to the external hard disk, and noted the time taken for each transfer. Similarly, for noting sequential speeds, we transferred a single 1 GB file.

Price Index

Price Index was calculated using the price-per-MB ratio. That is the ratio of the actual usable space—not the rated space, as there is always some loss due to formatting—and the price of the hard drive.

How The Awards Were Given

The score from Features, Build Quality, Performance and Price Index are given weightages relevant to the specific category. An overall score out of 100 is then calculated. The product that scores highest here is adjudged the winner of the Digit Best Buy Gold award for its category. The second-highest scorer gets the Digit Best Buy Silver award.